

FindIt

by

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Table 1: Document Update History

Date	Updates
10/02/2025	RD, RT, CV: Added Development Plan chapter.
09/30/2025	RD, RT, CV: - Added Bibliography to appendix - Moved Weekly Reports to appendix.
09/25/2025	RD, RT, CV: Added Weekly Report 4
09/18/2025	RD, RT, CV: - Edited and Finished Team Declaration chapter - Added Weekly Report 3
09/16/2025	RD, RT, CV: Added Team Declaration chapter
09/12/2025	RD, RT, CV: - Added Roy Tas's introduction - Added Weekly Report 2 - Created System overview diagram
09/04/2025	RD, RT, CV: - Added introduction page, introducing the team. - Added Weekly Report 1

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Chapter 1

Introduction

- Ryan Davis, Cory Vitanza, and Roy Tas

My name is Ryan Davis, and I am a Software Engineering major. I am in my senior year of study pursuing my bachelor's degree. I hope to continue my Software Engineering career one day as the founder of a startup company.

Hi, my name is Cory Vitanza, a 4th-year Software Engineering student also currently pursuing a Master's in Engineering Management. I've gained coding experience from my classes and real-world experience in QA and testing during my internship at Trimble MAPS. After my time here at Stevens, I plan to continue working in QA, hopefully in a management position soon enough. My favorite coding language is C++, and I love working on challenging projects. Outside of coding, I'm into music, cars, video games, and watching / playing sports, especially basketball and football

My name is Roy Tas I am a senior Engineering Management major concentrating in Financial Engineering. I have some C++ and Python experience but I will be focusing more on the scalability and logistics of our project. Post-graduation I want to pursue a career in jewelry manufacturing.

Chapter 2

Team Declaration

- Ryan Davis, Cory Vitanza, and Roy Tas

2.1 Team Information

- **Team Name:** FindIt
- **Team Members:**
 1. Ryan Davis
 2. Cory Vitanza
 3. Roy Tas

2.2 Project Proposal

Patrons with disabilities such as autism or conditions such as dementia tend to wander or get lost, causing families and caregivers of such patrons great amounts of stress. We are FindIt and we aim to mitigate stress, with our new tracking solution employing LoRaWAN technology to track locations at far distances, with extensive battery life.

Our device aims to demonstrate that a low-cost and accessible solution can balance affordability, reliability, and ease of use- as compared to current tracking solutions like AirTags or other commercial tracking devices. We envision FindIt as a scalable option that extends beyond families to institutional applications such as nursing homes, group homes, and schools, where safety and monitoring are critical. Our device comforts families with efficient tracking tools to ensure comfort when their loved ones are lost.

2.3 Mission Statement

The mission of FindIt is to design and deliver a reliable, affordable, and user-friendly location tracking solution for families and caregivers of individuals at risk of wandering. Our goal is to reduce stress by providing a reliable tool that allows quick and efficient tracking in critical situations. The business need is the lack of affordable long-range tracking technology for families,

and the business intent is to bridge that gap by leveraging LoRaWAN for extended battery life and coverage.

2.4 Key Drivers

Several key drivers motivate the development of this solution.

There is a great need for improved tracking solutions for wandering patrons. Wandering is prevalent among groups with autism or dementia, and existing solutions are expensive, insufficient, and at times inaccurate. We aim to minimize these issues with the development of our device.

Current solutions for asset tracking, like AirTags or cellular-based trackers, can become expensive for long-term usage. Using LoRaWAN protocols, our device reduces recurring costs and enhances battery life.

Technological accessibility is also an important motivator, as our group aims to reduce the stress of figuring out an application when time is of the essence. Our service is also subscription-free, in order to benefit all those in need.

LoRaWAN's versatility allows for uses beyond families. Group facilities such as schools, assisted living centers, or hospitals could benefit from a low-cost, universal system to monitor the safety of many individuals simultaneously.

2.5 Key Constraints

- **Network Coverage:** LoRaWAN's effectiveness depends on gateway proximity. Limited coverage in some areas may impact reliability, requiring design options for sparse networks. LoRaWAN coverage maps detail that Urban areas, specifically in the Northeast, have excellent coverage.
- **Battery Life vs. Device Size:** Maintaining strong battery life while preserving the small size of this device is optimal for the constant carriage of our unit. The device must remain lightweight and comfortable for continuous use.
- **Privacy and Security:** Storing and sending location data leads to privacy concerns. Secure data handling, LoRaWAN encryption, and user-controlled permissions are necessary to maintain trust with the user.
- **Cost Targets:** Affordability is a primary selling point. Hardware, manufacturing, and deployment costs must remain competitive without sacrificing reliability.
- **Interface Simplicity:** Achieving a clean, intuitive interface while maintaining accessibility features is critical for the tracking unit to reach diverse user bases with differing technological expertise.

Chapter 3

Development Plan

- Ryan Davis, Cory Vitanza, and Roy Tas

3.1 Introduction

The purpose of this project is to design, implement, and validate a location-tracking and data communication system using LoRaWAN-enabled hardware integrated with a cloud-based software platform. The system aims to provide reliable long-range wireless data transmission between embedded devices and a backend service, with supporting web-based tools for monitoring and management. By combining embedded development, cloud infrastructure, and a user-facing interface, this project demonstrates the end-to-end development process of a distributed system.

Success criteria for the project include (1) establishing reliable LoRaWAN communication between the embedded board and backend, (2) ensuring data is properly stored, processed, and displayed through the frontend application, and (3) passing acceptance testing with the Wearable Robotics Systems Lab. From a user perspective, the system enables simple, reliable collection and visualization of data in real time, serving as a proof-of-concept platform that can be extended to future applications.

3.2 Roles and Responsibilities

Clear definition of roles and responsibilities ensures accountability and effective collaboration within the team. Since this project is being developed by a small group, several members will serve multiple roles across development, testing, documentation, and risk management. The following assignments identify primary responsibilities, while recognizing that all team members will contribute collaboratively to design decisions, implementation, and review processes.

1. Development Lead: Ryan
2. Buildmeister: Roy
3. Architect: Cory
4. Developers: Ryan (Backend), Cory (Frontend), Roy (Infrastructure)

5. Test Lead: Cory
6. Testers: All
7. Documentation: All
8. Documentation Editor: All
9. Designer: Cory
10. User advocate: Ryan Davis
11. Risk Management: Roy
12. System Administrator: Ryan
13. Modification Request Board: All
14. Requirements Resource: Roy
15. Customer Representative: All
16. Customer responsible for acceptance testing: Wearable Robotics Systems Lab

3.3 Method

These are unique to software development, although there may be some overlap.

3.3.1 Software

1. Language(s):
 - C (Arduino IDE)
 - Python (3.x)
 - TypeScript (React)
 - SQL (PostgreSQL via Supabase)
2. Operating System(s):
 - Arduino embedded OS (board-specific)
 - Linux/Ubuntu (for Docker and AWS deployment)
 - macOS/Windows (development environments)
3. Software packages/libraries used:
 - LoRaWAN protocol (TCP/IP stack integration)
 - Supabase (PostgreSQL backend service)

- Google Maps API
- React (TypeScript frontend framework)
- Docker (containerization)
- AWS (deployment and hosting)
- GitHub (version control)
- Kanban Board (Agile task management)

4. Code conventions:

- GitHub repository guidelines for commits and pull requests
- ESLint + Prettier for TypeScript (frontend)
- PEP 8 for Python (backend tools)
- Arduino C style conventions

3.3.2 Hardware

1. Development Hardware

- Wio Tracker 1110 Development Board
- Semtech LR1110 LoRa® transceiver

2. Test Hardware

- Wio Tracker 1110 Development Board
- LoRaWAN Gateway (connected to The Things Stack)

3. Target/Deployment Hardware

- Wio Tracker 1110 Development Board
- Semtech LR1110 LoRa® transceiver
- LoRaWAN Gateway (The Things Stack integration for live deployment)

3.3.3 Backup plan (individual and project)

In case the Wio Tracker is not capable of the deployment of our software, the FindIt team will need to research and develop board alternatives. One option for this scenario would be to use a Raspberry Pi Nano, implemented with an external LoRaWAN module to allow for data transmission. Even with such addition, it is necessary for our unit to maintain battery consumption amongst boards, ensuring our unit has extensive battery life for users.

3.3.4 Review Process

1. Will you do architecture, usability, design, security, privacy or code reviews?
 - Code reviews will be completed upon the completion of Agile tasks on our Kanban board.
2. What approach will you use for the reviews (formal, informal, corporate standard)?
 - Informal reviews will be employed to achieve the approval of all members of the team, ensuring that all cases are handled in new implementation.
3. Who is responsible for the reviews and resolving any issues uncovered by the reviews?
 - Upon the implementation of a component, the other two members of the team will perform a code review. Any issue uncovered from code review will be resolved by the member working on the component.

3.3.5 Build Plan

1. Revision control system and repository used
 - GitHub (Version Control)
2. Continuous integration
 - Circle CI
3. Deadlines for the builds
 - Depending on our planned sprints, our Kanban board will present deadlines based on tasks.
4. Regression test process – see test plan

3.3.6 Modification Request Process

1. MR tool: For modification requests, we will use our Kanban board on GitHub to issue any modification requests as a task in the board.
2. Decision process: Decisions for modification will be decided during heartbeat meetings, after a discussion among the team concerning the modification. The team will proceed to assign a task to the Kanban board on the methodology and developers necessary for the task.
3. State whether there will be two process streams one during development and one after development:
 - During development, there will be two process streams. One is this programming of the development board and the other will be for the front-end user interface. The two streams will be integrated for deployment.

3.4 Virtual and Real Workspace

3.4.1 Virtual Workspace

Our team will utilize texts for direct communication, as well as a Discord server used for the sharing of files, resources, and communication. GitHub will also be used for secondary communication, implementing comments and changes to the code base via commits and merge requests.

3.4.2 Real Workspace

The team's regular workspace will be the Software Engineering Lab, primarily for heartbeat meetings and the development of our program. Meetings with Dr. Zanotto will be held in his office or over Zoom, when necessary.

3.5 COMMUNICATION PLAN

3.5.1 Heartbeat Meetings

Heartbeat meetings will be held at least twice per week with all group members present. These meetings serve to take the “pulse” of the project by providing quick updates, raising blockers, and reviewing open issues and risks. Each meeting will follow a structured agenda that includes:

- Individual progress updates
- Discussion of current blockers or risks
- Review of action items and open issues

Notes from each meeting will be recorded in the project document, and the GitHub Kanban board will be updated accordingly. These meetings are designed to be short, ideally during our allotted class-time.

3.5.2 Status Meetings

Status meetings will occur once a month and will include the project group and Dr. Damiano Zanotto. The purpose of these meetings is to present overall project progress, demonstrate completed work, and receive feedback on areas for improvement. If issues arise during these meetings, they will be addressed separately in an issues meeting. These meetings are intended to be concise, focusing on high-level updates and progress checkpoints.

3.5.3 Issues Meetings

Issues meetings will be scheduled whenever significant problems arise that require group consensus or immediate attention. Alerts for these meetings will typically be raised through a Discord message. If consensus is reached that a meeting is necessary, it will be scheduled as soon as all

group members are available. These meetings provide a dedicated space for problem-solving and decision-making outside of the regular heartbeat meetings.

3.6 Timeline and Milestones

The following milestones outline the major deliverables for the project. Each milestone represents a 100% complete item, with critical participants, begin and end dates, and deliverables clearly defined.

- **Milestone 1: Project Setup & Requirements Finalization**
 - Begin: Oct 3, 2025 End: Oct 17, 2025
 - Deliverables: Project repository initialized, Kanban board configured, requirements document finalized, hardware/software resources received and confirmed.
- **Milestone 2: Core System Architecture & Prototype Communication**
 - Begin: Oct 17, 2025 End: Oct 31, 2025
 - Deliverables: Draft system architecture, initial Arduino board programming for basic LoRaWAN communication, mock frontend interface, database schema in Supabase.
- **Milestone 3: Integration of Hardware and Software Components**
 - Begin: Oct 31, 2025 End: Nov 14, 2025
 - Deliverables: Hardware successfully transmitting data to backend, frontend connected to database, Docker setup for local deployment, integration testing initiated.
- **Milestone 4: Feature Implementation & System Testing**
 - Begin: Nov 14, 2025 End: Nov 28, 2025
 - Deliverables: Full set of planned features implemented, CI/CD pipeline operational, unit + integration tests passing, preliminary user documentation drafted.
- **Milestone 5: Final Review & Delivery**
 - Begin: Nov 28, 2025 End: Dec 12, 2025
 - Deliverables: Final acceptance testing completed, documentation finalized, project demonstration delivered, final repository and deployment package prepared for submission.

This document will be re-issued at each milestone to reflect progress and highlight changes since the previous revision.

3.7 Testing Policy/Plan

Our testing policy combines automated and manual approaches to ensure that the project meets functional requirements and maintains high reliability. The following types of testing will be performed:

- **Unit Testing:** We will implement unit tests for critical components, including geofence logic and location-based features. Pytest will be the primary testing framework. Unit testing will begin as soon as core modules are developed and will continue throughout the project.
- **Integration Testing:** Integration testing will verify interactions between different system components, such as API calls to The Things Stack and web service calls to the Google Maps API. These tests will be mocked where appropriate to ensure reliability and repeatability.
- **System Testing:** End-to-end system testing will be conducted once a minimum viable product is available. This will include route pings, location checking, and verifying that hosted meals can be created, displayed, and joined.
- **Acceptance Testing:** Acceptance testing will be performed in collaboration with Dr. Damiano Zanotto. This testing will validate that the system meets the agreed project requirements and is ready for delivery.
- **Regression Testing:** Regression tests will be rerun whenever bug fixes or new features are introduced to ensure that existing functionality remains stable.
- **Testing for Quality Attributes:** While performance and security are not the primary focus, geolocation accuracy and reliability will be emphasized. Specific attention will be given to ensuring that geofences trigger consistently and API interactions are fault-tolerant.
- **Continuous Integration (CI):** CircleCI will be used to automate testing during development. All code changes will trigger automated test runs to catch issues early.

Our stopping criteria for testing is defined as achieving at least 80% unit test coverage across critical modules, successful integration of geofencing and API functionality, and passing acceptance testing with project stakeholders.

3.8 Risks

Risks that we must mitigate throughout the development of our product include various issues on all scales of development. With the development board, we want to ensure that the battery life is competitive with existing tracking units, such as AirTag. We also would like to ensure that the tracking unit is able to provide precise location data for the user to accurately locate the unit.

3.9 Assumptions (required)

For the development of our program, it is assumed that:

- Publicly accessible LoRaWAN gateways populate our region, allowing for fluid connection between our board to LoRaWAN network servers.
- End users have mobile devices or computers that are able to access our website, providing them access to their tracking units.

3.10 DISTRIBUTION LIST

This document will be shared with the members of the FindIt team, as well as Dr. David Darian Muresan and Dr. Damiano Zanotto throughout our development processes.

3.11 IRB Protocol

Our project does not require IRB protocol.

3.12 Required Resource and Budget

3.12.1 Hardware Resources

For our tracking unit, the only development expense will include the Wio Tracker 1110 development board, costing \$30.

3.12.2 Software Resources

Depending on the size of our infrastructure, deployment on AWS as well as the storage of historical data on Supabase will require subscriptions in order to maintain the usage of our software.

3.13 Use Cases

Table 3.1: Initial Use Cases

Use Case ID	Description
UC1	Geofence Alert – The system detects when the tracker exits a predefined boundary and sends an alert to the caregiver’s phone.
UC2	Multi-Resident Monitoring – A caregiver monitors multiple trackers simultaneously and receives notifications when any resident leaves a safe area.
UC3	Battery Status Alert – The tracker measures its remaining battery and sends a notification when power is low.
UC4	Cost-Efficient Deployment – The system operates without subscription costs, relying on LoRaWAN gateways, and supports scaling across many patients.
UC5	Emergency Alert – A user can trigger an alert in case of a fall or emergency, notifying caregivers or family members for immediate response.

3.14 Requirements Mapped to Use Cases

Table 3.2: Requirements Mapping to Use Cases

Use Case ID	Requirement	Type	Priority
UC1	The system shall support geofencing with alerts when outside boundaries.	Functional	Must
UC1	The system shall provide real-time location updates via mobile app.	Functional	Must
UC2	The system shall support multi-device monitoring.	Functional	Must
UC2	The system shall provide configurable notifications for multiple users/devices.	Functional	Should
UC3	The device shall monitor battery level and send alerts below threshold.	Functional	Must
UC3	The system shall provide push notifications for device status (battery, connectivity).	Functional	Should
UC3	The system shall maintain battery life for over a year (competing with AirTag battery life).	Functional	Could
UC4	The system shall operate without subscription costs.	Constraint	Must
UC4	The system shall scale to support multiple concurrent devices.	Non-Functional	Should
UC5	The system shall have a button to notify users of the interface that an emergency is occurring.	Functional	Should
UC5	The button could transmit Morse code data.	Functional	Could

Chapter 4

Requirements

– *Author Name*

4.1 Stakeholders

- Wearable Robotics Systems Lab
- Assisted Living Homes
-

4.1.1 Customers

Clients and users. Who are they, why do they use your system

4.1.2 Sponsors

4.1.3 Engineering and Technical Persons

4.1.4 Regulators

4.1.5 Third Parties

4.1.6 Competitors

Systems which provide similar functions.

4.2 Key Concepts

List the key concepts and their definitions that relate to your project. These concepts and terms should be used consistently throughout this document and in your project. This can point out to the Glossary chapter.

4.3 User Requirements

Abstract statements written in natural language with accompanying informal diagrams. You may use user stories or simple use case diagram(s). The user requirements should be numbered and linked to use-cases and user-stories. Here is an example.

Table 4.1: User Requirements Table

Requirement	Priority	Use Case(s)
Quality Requirement 1 (reqqFirstQualityRequirement) <i>The system shall perform the task in less than one second.</i> • Need: <i>explain the need.</i> • Stability: <i>explain.</i>	MustHave	UC??
Functional Requirement 1 (reqffFirstFunctionalRequirement) <i>The system shall perform the task of adding 1+1.</i>	ShouldHave	UC??
Constraint Requirement 1 (reqcFirstConstraintRequirement) <i>The system shall not interfere with local Wi-Fi communication channels.</i>	CouldHave	UC??
Interface Requirement 1 (reqiFirstInterfaceRequirement) <i>The system shall have a C++ API.</i>	WouldHave	UC??
Business Requirement 1 (reqbFirstBusinessRequirement) <i>The system shall be FDA approved.</i>	WouldHave	UC??

At some point in your document, you will need to define your use case. For this to all work, you also need to define newtheorem for the requirements and use cases, in your root manual.tex.

4.4 System (Constraints) Requirements

More detailed descriptions of the services and constraints from the perspective of the system to meet the user requirements. Should be structured and precise. More detailed that describe the user requirements and focus on the basic flow, alternative flow, exceptions, pre-conditions, and post-conditions, and special requirements for each abstract user requirements. May use use- case template for this. System requirements should be numbered and traced back to the user requirements.

4.5 Non-functional (Quality) Requirements

Any important non-functional requirements for your project. These would include any performance and quality requirements (i.e. numbers, such as bandwidth, time, speed, rates, etc.). Can add here any type of a multitude of quality requirements that define, in large part, the system architecture.

4.6 Domain (Business) Requirements

Business rules and regulations that impact what the system does.

4.7 Other Requirement Types

If there is a reason to define a new, non-overlapping requirement class, you can create more requirement classes as needed, but be careful to not create too many.

Appendix A

Weekly Reports

- Ryan Davis, Cory Vitanza, and Roy Tas

A.1 Week Report 5 (10/02/2025)

A.1.1 Progress Made

This week, the team completed and presented our Demo, which covered our Project Mission and Development Plan. We received constructive feedback from professors and peers, which will help us refine our development strategy. In addition, the team set up our GitHub repository and established a project Kanban board within GitHub Projects to track tasks and streamline collaboration in an Agile workflow.

A.1.2 Next Week's Plan

For next week, the team plans to order the new development board and begin work on multiple components of our system. This includes initiating development of the frontend and user interface, setting up the Supabase database for data storage, and integrating CircleCI for continuous integration with our GitHub repository. We will also begin configuring our DevOps infrastructure to support efficient development and deployment workflows.

A.2 Week Report 4 (09/25/2025)

A.2.1 Progress made

This week, our team has decided that we have to purchase a new board. This new board will remove Amazon Sidewalk capabilities, allowing us to narrow our focus on the LoRaWAN capabilities of the development board. Unfortunately, our current board's bootloader is configured to the wrong board-preset, thus preventing us from accessing certain areas of our current board. Our current board thinks its a Seeed XIAO board, but in reality, it is a WIO Tracker 1110 development board. This change will further allow us to build our tracking unit.

A.2.2 Next Week's Plan

Next week, the team plans to complete our 1st presentation, highlighting our project mission and development plan for this technology stack. We also plan to acquire the new board, start creating our GitHub, along with the agile board and task assignments for our members. We will begin to start developing the board to receive gps data, and establish our pipeline for data transmission.

A.2.3 Risk Mitigation

As we implement our tracking algorithm into the development board, a risk that we would like to minimize is the accuracy of the tracking data. We aim to implement indoor and outdoor tracking data, allowing for improved precision in location data.

A.3 Week Report 3 (09/18/2025)

A.3.1 Progress made

This week, the team was able to compose our [Team Declaration](#), which includes an outline of our proposed project, our mission, key drivers for our design, and key constraints to keep in mind throughout our development.

A.3.2 Next Week's Plan

Next week, the team plans to debug the current Seeed Studio LoRaWAN Arduino device, through configuring the bootloader to ensure complete functionality of the board, including the on-board buttons, GPS module, LoRA module, and the serial connection. The team will also begin to compose our project mission and development plan for the upcoming presentation.

A.3.3 Risk Mitigation

The risk that the team is currently facing coincides with the debugging of the LoRaWAN device. Currently, the device is flashed with print statements to the serial monitor, but is constrained to this operation without access to other board features, such as button usage and the ability to flash the device again. To minimize this risk, we are looking into alternative devices, such as a Raspberry Pi Nano, but will also continue to attempt to debug the current Seeed Studio device.

A.4 Week Report 2 (09/12/2025)

A.4.1 Progress Made

This past week, the team created a deployment diagram that illustrates the system's interfaces and necessary developments. The team also met with Professor Damiano Zanutto to further expand and outline the scope of our asset tracking project. We are moving in the direction of developing this low-bandwidth and inexpensive tracker for people with autism or dementia patients who may wander, allowing caregivers and family members to locate the tracker. We are also looking to host the data locally, directly from the machine using The Things Stack, which is a LoRaWAN network server that allows API retrieval of data. We would also

like to look into implementing a machine learning model which can process GPS data for precise indoor and outdoor geo-location tracking.

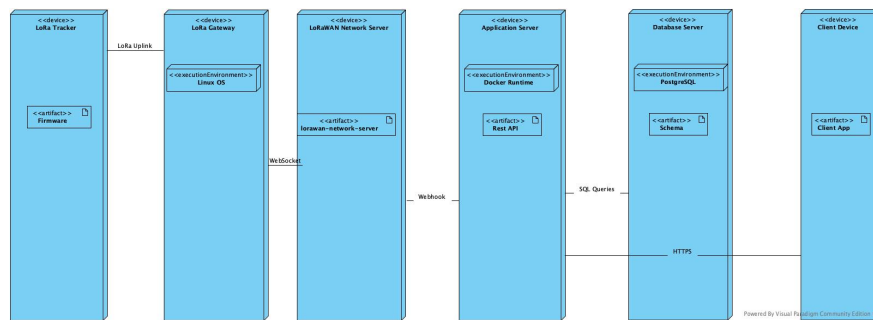


Figure A.1: System Diagram

A.4.2 Next Week's Plan

For next week, the team has split the planning of our development into different components. Ryan's research will involve looking into open source tracking projects that can be tailored towards our model. Roy's research will look into the scalability of our model, and the extent to which our system will be built out, and Cory's research will explore how our model can avoid using services like AWS to minimize the transactions of our data. We will also continue to debug the LoRaWAN machine that we are working with, and will also explore our other options with such devices.

A.5 Week Report 1 (09/05/2025)

A.5.1 Progress made

Our group which currently consists of 2 members, Ryan Davis and Cory Vitanza, are currently exploring our options regarding our project. We have reached out to classmates and professors and plan on making our decision within the next week.

A.5.2 Next Week's Plan

Following my Summer Research with Doctor Damiano Zanotto, I plan on requesting my Summer research project to be extended throughout the course of my senior design project. This project includes asset tracking using embedded minicomputers communicating over a LoRaWAN connection in order to process gps data.

A.5.3 Backup Plan

If this plan does not work in our favor, we plan to reverse engineer and develop a "magic mirror", which acts as a dashboard, containing different modules showing a handful of information. This is driven by a raspberry pi board over an internet connection, further displayed on a monitor and a 1-way mirror.

Glossary

CouldHave This defines the third highest priority requirement. The system could implement all of the tasks, requirements, or anything that is marked this way, but if resources are limited, it can be left out of the current and next version. Build in two versions from now. [15](#)

MustHave This defines the first highest priority requirement. All of the tasks, requirements, or anything that is marked this way are build in the current version. [15](#)

ShouldHave This defines the second highest priority requirement. The system should implement all of the tasks, requirements, or anything that is marked this way, but if resources are limited, it can be left out of the current version. Build in next version. [15](#)

WouldHave This defines the lowest priority requirement. The system would like to implement all of the tasks, requirements, or anything that is marked this way, but only if resources are available. It can be left out of all future versions. [15](#)

Bibliography

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